Optical turbidity sensing

ENGR 130 · Module 2

Experiment overview

Collected optical-sensor voltages for turbidity standards, fitted a nonlinear calibration equation and developed an algorithm for repeated sampling and a stop condition. One report lists a 1.55% comparison error for a 250 NTU check.

Variables and instrumentation

Sensor voltage, turbidity, repeated sampling, threshold logic

Analysis and presentation

Calibration curve; predicted turbidity; decision logic demo

Engineering relevance

Connects instrumentation with automated quality checks and programmable control decisions.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1jyxetY9Sn2BWC5ysGGjCd8Kz5wL51f36mjG3-4eT-uk/edit